

Ar/O2 Global Model

Reference - C. Lee and M. A. Lieberman: Global model of Ar, O₂, Cl₂, and Ar/O₂ high density plasma discharges

1. Neutral Gas Density Formula

To establish the base neutral species concentration from the input pressure, the ideal gas law is applied at a steady-state ambient neutral gas temperature.

Formula:

$$n_g = p_{Pa} / (k_B * T_{neutral})$$

Where:

- n_g : Total neutral gas number density (particles per cubic meter)
- p_{Pa} : Reactor pressure converted to Pascals (computed from pressure in mTorr via $p_{Pa} = p_{mTorr} * 0.1333$)
- k_B : Boltzmann constant (1.38065×10^{-23} Joules per Kelvin)
- $T_{neutral}$: Background neutral gas temperature, set to a constant 600 Kelvin to account for plasma-induced heating of the neutral background

2. Ion Mean Free Path and Cross-Sections

The average distance an ion travels before colliding with a background neutral atom/molecule determines the transport and diffusion regime of the plasma.

Formula:

$$\lambda_{i} = 1 / (\sqrt{2} * \sigma_{cross} * n_g)$$

Where:

- λ_i : Ion-neutral mean free path (meters)
- σ_{cross} : Effective total collision cross-section for ion-neutral interactions, taken as a representative value of 4.0×10^{-19} square meters for oxygen-dominated mixtures
- n_g : Total neutral gas number density calculated in Equation 1

3. Characteristic Velocity Scales (Bohm Velocity)

Ions escaping the core plasma must cross the plasma-sheath boundary. The velocity at which they enter this boundary is governed by the electron temperature.

Formula:

$$u_{B_x} = \sqrt{e_{charge} * T_e / M_x}$$

Where:

- u_{B_x} : Bohm velocity of ion species x (meters per second), where x represents Argon ions (Ar⁺), molecular oxygen ions (O₂⁺), or atomic oxygen ions (O⁺)
- e_{charge} : Elementary charge of an electron (1.602×10^{-19} Coulombs)
- T_e : Electron temperature (electron-volts, eV)

- M_x : Mass of the respective ion species in kilograms ($M_{Ar} = 39.948$ amu, $M_{O2} = 31.998$ amu, $M_O = 15.999$ amu; all scaled by the atomic mass unit $m_u = 1.66054 \times 10^{-27}$ kg)

4. Electronegative Spatial Scaling (h-Factors)

Because negative ions pull electrons inward, the flat spatial profile of a pure Argon discharge transitions into a parabolic profile when negative ions are present. The h-factors scale the bulk core density to the boundary sheath density.

Formulas:

$$hL_{ep} = 0.86 / \sqrt{3 + L / (2 * \lambda_i) + (0.86 * L * u_B / (\pi * Da))^2}$$

$$hR_{ep} = 0.8 / \sqrt{4 + R / \lambda_i + (0.8 * R * u_B / (2.405 * J_{1_2405} * Da))^2}$$

Modifications for Electronegativity (when α is greater than 0.05):

$$hL_{en} = hL_{ep} / \sqrt{1 + \alpha}$$

$$hR_{en} = hR_{ep} / \sqrt{1 + \alpha}$$

Where:

- hL_{ep} , hR_{ep} : Axial (L) and Radial (R) sheath-to-bulk density ratios under electropositive conditions
- hL_{en} , hR_{en} : Axial and Radial sheath-to-bulk density ratios adjusted for negative ion profiling
- L: Cylinder length of the reactor chamber (0.075 meters) *as taken in paper*
- R: Cylinder radius of the reactor chamber (0.1525 meters)
- α : Electronegativity ratio, defined explicitly as the negative ion density divided by electron density (n_{O_m} / n_e)
- Da: Ambipolar diffusion coefficient, calculated via $Da = (e_{charge} * T_e) / (M_{O2} * \nu_{in})$, where ν_{in} is the ion-neutral collision frequency given by u_B / λ_i
- J_{1_2405} : The value of the first-order Bessel function evaluated at its first root

5. Effective Loss Area Matrix

The geometric area through which positive ions are actively lost to the reactor walls is modified by the directional h-factors.

Formula:

$$A_{eff} = 2 * \pi * R^2 * hL + 2 * \pi * R * L * hR$$

Where:

- A_{eff} : Total effective area for ion loss to the surfaces (square meters)
- R: Reactor radius (meters)
- L: Reactor length (meters)
- hL , hR : Spatial scaling ratios calculated from Equation 4 (using either electropositive or electronegative variants depending on the local α value)

6. Atomic Oxygen Wall Recombination Loss

Dissociated oxygen atoms don't remain atomic forever; they hit the reactor walls and recombine into stable molecular oxygen molecules based on a surface probability factor.

Formula:

$$k_{loss_O_wall} = (GAMMA_{REC} * v_{th_O} * A_{total}) / (4 * V)$$

Where:

- $k_{\text{loss_O_wall}}$: Volumetric rate constant for atomic oxygen loss to the walls (1 per second)
- GAMMA_REC : Surface recombination coefficient, set to 0.01 (representing the probability that an O atom striking a clean quartz or glass wall will stick and form O₂)
- A_{total} : Total physical area of the reactor inner surfaces, given by $2\pi RL + 2\pi R^2$ (square meters)
- V : Total physical volume of the reactor chamber, given by $\pi R^2 \cdot L$ (cubic meters)
- $v_{\text{th_O}}$: Thermal velocity of atomic oxygen, computed using the neutral gas temperature via $v_{\text{th_O}} = \sqrt{8 \cdot k_B \cdot T_{\text{neutral}} / (\pi \cdot M_{\text{O}})}$

TABLE VI. Argon–oxygen reactions.

Reaction	Rate coefficients	Reference
$\text{O}^- + \text{Ar}^+ \longrightarrow \text{Ar} + \text{O}$	$k = 2.70 \times 10^{-7} \text{ cm}^3 \text{ s}^{-1}$	a
$\text{O}_2 + \text{Ar}^* \longrightarrow \text{Ar} + \text{O}_2$	$k = 1.12 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$	b
$\text{O} + \text{Ar}^* \longrightarrow \text{Ar} + \text{O}$	$k = 8.10 \times 10^{-12} \text{ cm}^3 \text{ s}^{-1}$	b
$\text{O}_2 + \text{Ar}^* \longrightarrow \text{Ar} + \text{O} + \text{O}$	$k = 5.80 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$	c
$\text{O}_2 + \text{Ar}^+ \longrightarrow \text{O}_2^+ + \text{Ar}$	$k = 1.20 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$	38
$\text{O} + \text{Ar}^+ \longrightarrow \text{O}^+ + \text{Ar}$	$k = 1.20 \times 10^{-11} \text{ cm}^3 \text{ s}^{-1}$	d

^aThe reaction rate was assumed to be identical to the O^-/O^+ recombination rate.

^bEstimated from similar quenching reactions for Cl₂ and N₂ given in Refs. 13 and 14.

^cEstimated from similar reactions in NH₃ given in Ref. 39.

^dEstimated to be the same as the O_2/Ar^+ reaction obtained from Ref. 38.

7. Species Mass Balance Residual Equations

The core solver determines steady-state values by driving the difference between a species' local production rate and its destruction/loss rate to exactly zero. All rate coefficients (k_{iz} , k_{exc} , etc.) are evaluated as functions of electron temperature (T_e) using the paper's analytical exponential fits.

Argon Metastable (Ar*) Balance:

$$\begin{aligned} \text{res_Arst} = & k_{\text{exc_Ar}} \cdot n_{\text{Ar}} \cdot n_e - k_{\text{iz_Arst}} \cdot n_{\text{Arst}} \cdot n_e - k_{\text{de_Arst}} \cdot n_{\text{Arst}} \cdot n_e - 2 \cdot \\ & k_{\text{pool_Ar}} \cdot n_{\text{Arst}}^2 - k_{\text{q_Arst_O2}} \cdot n_{\text{Arst}} \cdot n_{\text{O2}} - k_{\text{q_Arst_O}} \cdot n_{\text{Arst}} \cdot n_{\text{O}} - \\ & k_{\text{dis_Arst_O2}} \cdot n_{\text{Arst}} \cdot n_{\text{O2}} \end{aligned}$$

Atomic Oxygen (O) Balance:

$$\begin{aligned} \text{res_O} = & 2 \cdot k_{\text{dis_O2}} \cdot n_{\text{O2}} \cdot n_e + k_{\text{att_O2}} \cdot n_{\text{O2}} \cdot n_e + 2 \cdot k_{\text{dis_Arst_O2}} \cdot n_{\text{Arst}} \cdot n_{\text{O2}} \\ & - k_{\text{iz_O}} \cdot n_{\text{O}} \cdot n_e - k_{\text{cx_Arp_O}} \cdot n_{\text{Ar_p}} \cdot n_{\text{O}} - k_{\text{det_O}} \cdot n_{\text{O_m}} \cdot n_{\text{O}} - k_{\text{loss_O_wall}} \cdot \\ & n_{\text{O}} \end{aligned}$$

Argon Ion (Ar+) Balance:

$$\begin{aligned} \text{res_Ar_p} = & k_{\text{iz_Ar}} \cdot n_{\text{Ar}} \cdot n_e + k_{\text{iz_Arst}} \cdot n_{\text{Arst}} \cdot n_e + k_{\text{pool_Ar}} \cdot n_{\text{Arst}}^2 - k_{\text{cx_Arp_O2}} \\ & \cdot n_{\text{Ar_p}} \cdot n_{\text{O2}} - k_{\text{cx_Arp_O}} \cdot n_{\text{Ar_p}} \cdot n_{\text{O}} - k_{\text{rec_O_Ar}} \cdot n_{\text{Ar_p}} \cdot n_{\text{O_m}} - (A_{\text{eff}} \cdot uB_{\text{Ar}} / \\ & V) \cdot n_{\text{Ar_p}} \end{aligned}$$

Molecular Oxygen Ion (O2+) Balance:

$$\text{res_O2_p} = k_{\text{iz_O2}} * n_{\text{O2}} * n_e + k_{\text{cx_Arp_O2}} * n_{\text{Ar_p}} * n_{\text{O2}} - k_{\text{rec_O_O2p}} * n_{\text{O2_p}} * n_{\text{O_m}} - (A_{\text{eff}} * u_{\text{B_O2}} / V) * n_{\text{O2_p}}$$

Atomic Oxygen Positive Ion (O+) Balance:

$$\text{res_O_p} = k_{\text{iz_O}} * n_{\text{O}} * n_e + k_{\text{cx_Arp_O}} * n_{\text{Ar_p}} * n_{\text{O}} - k_{\text{rec_O_Op}} * n_{\text{O_p}} * n_{\text{O_m}} - (A_{\text{eff}} * u_{\text{B_O}} / V) * n_{\text{O_p}}$$

Atomic Oxygen Negative Ion (O-) Balance:

$$\text{res_O_m} = k_{\text{att_O2}} * n_{\text{O2}} * n_e - k_{\text{det_O}} * n_{\text{O_m}} * n_{\text{O}} - n_{\text{O_m}} * (k_{\text{rec_O_Ar}} * n_{\text{Ar_p}} + k_{\text{rec_O_O2p}} * n_{\text{O2_p}} + k_{\text{rec_O_Op}} * n_{\text{O_p}})$$

Where:

- n_e : Electron number density (particles per cubic meter)
- n_{Ar} , n_{O2} : Ground state neutral densities
- n_{Arst} : Argon metastable density
- n_{O} : Dissociated atomic oxygen neutral density
- $n_{\text{Ar_p}}$, $n_{\text{O2_p}}$, $n_{\text{O_p}}$: Respective positive ion number densities
- $n_{\text{O_m}}$: Negative ion (O-) number density
- Terms ending in / V: Account for physical particle escape to the walls distributed across the active volume

8. Global Charge Neutrality Closure Relation

This relation acts as a foundational plasma physics constraint. Total negative charge carriers must equal total positive charge carriers.

Formula:

$$n_e + n_{\text{O_m}} = n_{\text{Ar_p}} + n_{\text{O2_p}} + n_{\text{O_p}}$$

In the explicit closure modification, this algebraic constraint is rearranged to calculate electron density directly:

$$n_e = (n_{\text{Ar_p}} + n_{\text{O2_p}} + n_{\text{O_p}}) - n_{\text{O_m}}$$

Where:

- n_e : Electron density
- $n_{\text{O_m}}$: Negative ion density
- $n_{\text{Ar_p}}$, $n_{\text{O2_p}}$, $n_{\text{O_p}}$: Positive ion densities

9. Total System Power Balance Matrix

The total high-frequency RF power absorbed by the plasma discharge must exactly match the sum of power lost through volume collisions (ionization, excitation, dissociation) and the kinetic energy carried away by particles striking the walls.

Formula:

$$P_{\text{abs_W}} = P_{\text{loss_vol}} + P_{\text{loss_walls}}$$

Where:

Volume Loss Component:

$$P_{\text{loss_vol}} = e_{\text{charge}} * V * n_e * (k_{\text{iz_Ar}} * n_{\text{Ar}} * 15.76 + k_{\text{exc_Ar}} * n_{\text{Ar}} * 11.56 + k_{\text{iz_O2}} * n_{\text{O2}} * 12.0 + k_{\text{dis_O2}} * n_{\text{O2}} * 5.6 + k_{\text{att_O2}} * n_{\text{O2}} * 4.0 + k_{\text{iz_O}} * n_{\text{O}} * 13.6)$$

Surface Sheath Loss Component:

$$P_{\text{loss_walls}} = e_{\text{charge}} * A_{\text{eff}} * (n_{\text{Ar_p}} * u_{\text{B_Ar}} * (7.0 * T_e) + n_{\text{O2_p}} * u_{\text{B_O2}} * (7.0 * T_e) + n_{\text{O_p}} * u_{\text{B_O}} * (7.0 * T_e))$$

Where parameters match definitions above, and the scalar coefficients inside the power equations represent fixed energetic barriers:

- 15.76, 11.56, 12.0, 5.6, 4.0, 13.6: Specific cross-sectional energy thresholds (in electron-volts) required to complete each respective electron impact reaction.
- $7.0 * T_e$: The total kinetic energy lost per electron-ion pair entering the plasma sheath. It accounts for the average kinetic energy of the escaping bulk thermal electrons entering the potential barrier ($2.0 * T_e$) combined with the kinetic energy gained by the positive ions accelerating through the plasma sheath drop to the wall ($5.0 * T_e$).

Based on the foundational paper by C. Lee and M. A. Lieberman ("Global model of Ar, O₂, Cl₂, and Ar/O₂ high-density plasma discharges", JVST A 13, 1995), the selection of governing equations and chemical reactions is rooted in rigorous plasma physics, balancing volume chemistry with geometry-driven boundary losses.

Understanding why these specific equations and reactions were chosen highlights the physical mechanisms they capture.

1. Why the Equations Were Chosen (The Framework)

A 0D Global Model relies on volume-averaged conservation principles rather than spatial grids. In high-density discharges like Inductively Coupled Plasmas (ICPs) operating at lower pressures, the plasma bulk remains relatively uniform, while sharp gradients are confined near the wall sheaths. This justifies using two macro-balance equations:

A. Particle Balance (Continuity Equation)

For each particle species (electrons, neutral atoms, molecules, and individual ion types like Ar⁺, O₂⁺, O⁺, and O⁻), you must satisfy:

Generation Rate = Loss Rate

- **Why this form?** In the volume, species are generated or destroyed via electron impact reactions and heavy-particle collisions. However, charged particles also escape to the reactor walls.
- **The Role of Sheaths (h_L and h_R):** Ions do not hit the walls at their bulk density (n_i); they drop sharply to a sheath-edge density (n_{is}). The equations use edge-to-bulk interpolation factors (h_L and h_R). These factors account for the ambipolar diffusion rates under the influence of electronegative negative ions (O⁻), which physically trap themselves in the center of the discharge while pulling positive ions outward.

B. Electron Power Balance Equation

The overall power balance tracks how the radio-frequency energy absorbed by the plasma (P_{abs}) is spent:

$$P_{\text{abs}} = P_{\text{ev}} + P_{\text{iw}} + P_{\text{ew}}$$

- **Why this form?** Absorbed power must perfectly balance three loss paths:
 1. P_{ev} (Volume losses): Energy electrons transfer to neutrals via elastic/inelastic collisions.
 2. P_{iw} (Ion wall loss): Kinetic energy ions carry out as they accelerate through the plasma sheath via the Bohm velocity (u_B).
 3. P_{ew} (Electron wall loss): High-energy electrons that overcome the sheath potential barrier and escape, taking $2T_e$ of thermal energy with them.

2. Why the Reactions Were Chosen (The Chemistry Locus)

When moving from a simple pure Argon discharge to an Ar/O₂ blend, the chemistry grows complex. The reactions specified in Table I (Oxygen), Table III (Argon), and Table VI (Argon-Oxygen mixtures) capture several essential behaviors:

A. Multi-Step Ionization (Metastables)

- **Reactions:** Electron impact excitation to Argon metastables ($e + \text{Ar} \rightarrow \text{Ar}^* + e$) followed by ionization ($e + \text{Ar}^* \rightarrow \text{Ar}^+ + 2e$), alongside Ground State Oxygen atom excitation to metastables ($\text{O}(1D)$).
- **Why?** In a dense plasma, step-ionization from long-lived metastable states significantly lowers the average electron energy required to sustain the plasma, because ionizing an already-excited state takes much less energy than ionizing a ground-state atom from scratch.

B. The Electronegative Lifecycle (Attachment, Detachment, and Recombination)

- **Reactions:** Dissociative Attachment ($e + \text{O}_2 \rightarrow \text{O} + \text{O}^-$), Electron Detachment ($e + \text{O}^- \rightarrow \text{O} + 2e$), and Ion-Ion Recombination ($\text{O}^- + \text{O}_2^+ \rightarrow \text{O} + \text{O}_2$ and $\text{O}^- + \text{O}^- \rightarrow 2\text{O}$).
- **Why?** Oxygen is an electronegative gas. Low-energy electrons attach to molecular oxygen to create O^- . This negative charge carrier cannot escape to the walls because of the positive plasma potential, so its population is strictly limited by volume electron-detachment and heavy ion-ion neutralizations.

C. Dissociation Balance (The Molecular Sink)

- **Reactions:** Electron impact dissociation ($e + \text{O}_2 \rightarrow \text{O} + \text{O} + e$) balanced by Surface Wall Recombination ($\text{O} \rightarrow \text{wall} \rightarrow 1/2 \text{O}_2$).
- **Why?** High absorbed powers heavily dissociate O₂ into atomic oxygen (O). This directly reduces the baseline density of O₂ molecules, effectively shrinking the "source" available for negative ion formation. The inclusion of wall recombination ensures the model accurately predicts the atomic-to-molecular neutral gas ratio.

D. Inter-Species Cross-Talk (Table VI)

- **Reactions:** Mutual neutralization between species ($\text{O}^- + \text{Ar}^+ \rightarrow \text{Ar} + \text{O}$) and Penning/Metastable quenching ($\text{O}_2 + \text{Ar}^* \rightarrow \text{Ar} + \text{O} + \text{O}$).
- **Why?** When mixing Argon and Oxygen, Argon metastables (Ar^*) transfer their stored energy to dissolve oxygen molecules. Furthermore, Ar^+ ions provide an additional channel to destroy O^- via volume recombination.

Why Exclude Other Reactions?

Lee and Lieberman purposefully excluded reactions like **three-body neutral recombination** because at low processing pressures (under 100 mTorr), three particles colliding simultaneously is statistically rare, rendering their rate coefficients negligible. Similarly, **dissociative ionization** of molecular ions is omitted because its high threshold energy makes the probability low, and the high dissociation rate keeps the base molecular ion density low anyway.

By implementing this precise set of balances and cross-reactions, your sweep correctly simulated how the plasma shifts its dominant positive ion from Ar^+ to O^+ as pressure increases, and captured the exact pressure-driven electron energy cost turnover that occurs in electronegative discharges.